

# Step-by-Step Instructions

## 1 Check container in WSL

Open WSL and verify your container is running:

```
docker ps
```

Check the port inside WSL:

```
ss -tlnp | grep <port>  
# or  
sudo netstat -tlnp | grep <port>
```

Example:

```
ss -tlnp | grep 9999
```

You should see `0.0.0.0:9999` and `docker-proxy` listening.

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## 2 Get WSL2 IP address

```
wsl hostname -I
```

The first IP is the one Windows must connect to, e.g., `172.25.124.29`.

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## 3 Open PowerShell as Administrator

You need elevated privileges to set up the portproxy.

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## 4 Remove old portproxy rules

```
netsh interface portproxy delete v4tov4 listenport=9999 listenaddress=127.0.  
0.1  
netsh interface portproxy delete v4tov4 listenport=9999 listenaddress=0.0.0.0
```

Optional: verify rules:

```
netsh interface portproxy show all
```

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## 5 Add new portproxy

Use `${}` for variables with colons in strings:

```
$port = 9999
$wsl_ip = (wsl hostname -I).Split(" ")[0]

netsh interface portproxy add v4tov4 listenport=$port listenaddress=0.0.0.0
connectport=$port connectaddress=$wsl_ip
netsh interface portproxy add v4tov4 listenport=$port listenaddress=127.0.0.1
connectport=$port connectaddress=$wsl_ip

Write-Host "Proxy set: localhost:${port} -> ${wsl_ip}:${port}"
```

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## 6 Allow firewall access

```
netsh advfirewall firewall add rule name="WSL portproxy inbound $port" dir=in
action=allow protocol=TCP localport=$port
```

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## 7 Restart IP Helper service

```
Restart-Service iphlpsvc
```

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## 8 Test connection from Windows

```
curl http://localhost:9999
```

If it doesn't work, try connecting directly to the WSL IP:

```
curl http://${wsl_ip}:9999
```

## 9 Optional: Automate with PowerShell script

Create `wsl-port-forward.ps1`:

```
$port = 9999
$wsl_ip = (wsl hostname -I).Split(" ")[0]

netsh interface portproxy delete v4tov4 listenport=$port listenaddress=0.0.0.0
netsh interface portproxy delete v4tov4 listenport=$port listenaddress=127.0.0.1

netsh interface portproxy add v4tov4 listenport=$port listenaddress=0.0.0.0 connectport=$port connectaddress=$wsl_ip
netsh interface portproxy add v4tov4 listenport=$port listenaddress=127.0.0.1 connectport=$port connectaddress=$wsl_ip

netsh advfirewall firewall add rule name="WSL portproxy inbound $port" dir=in action=allow protocol=TCP localport=$port

Restart-Service iphlpsvc

Write-Host "Proxy set: localhost:${port} -> ${wsl_ip}:${port}"
```

Run **as Admin** whenever WSL2 IP changes:

```
.\wsl-port-forward.ps1
```

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## 10 Quick Reference

1. Verify container is running in WSL (`docker ps` / `ss -tlnp`).
2. Get WSL2 IP: `wsl hostname -I`.
3. Open PowerShell as Admin.
4. Delete old portproxy rules.
5. Add new rules with `${}` variables.
6. Allow firewall for port.
7. Restart `iphlpvc`.
8. Test with `curl localhost:<port>`.
9. Optional: save as script for next time.

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**Tip:** The key things that usually break are: - WSL2 gets a new IP after reboot. - Portproxy rules reference old IP. - Firewall blocks inbound connections. - `iphlpvc` needs restart for rules to take effect.